



Network Tariff Report 2004

1st March 2004

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1 INTRODUCTION

As part of its *Electricity Distribution Price Determination*, the Office of the Regulator General (the Office), now known as the Essential Services Commission (the Commission) required the Distributors prepare an annual report on their network tariffs. This requirement is detailed in clause 2.3.13 of *Electricity Distribution Price Determination - Volume II*.

2 OVERVIEW OF NETWORK TARIFF COMPONENTS

Network tariffs consist of two components:

- Distribution related charges (DUoS) to recover the regulated revenues for the distribution business; and
- Transmission related charges (TUoS) to recover the costs of transmission use of system services provided by Victorian Energy Networks Corporation (VENCorp), transmission connection services provided by SPI PowerNet, equalisation payments prescribed by regulation and inter distribution system charges.

These various components are described in detail in Sections 2.1 to 2.3 below.

2.1 *Distribution Use of System Charges (DUoS)*

Leading up to each five year regulatory period the Commission determines the revenue requirement for each distribution business over the next regulatory period. The Office has utilised a building block methodology to determine the amount of required revenue for the purposes of the previous price review. Recovery of the aggregate distribution revenue requirement determined by the Commission is achieved through the application of DUoS tariffs. In terms of distribution prices from 2001 to 2005, the final outcome of the Distribution Price Review for CitiPower was a real reduction in average distribution prices of 11.2% in 2001 and 1% in subsequent years. This 1% reduction is reflected in the approved 2004 tariffs.¹

The application of distribution tariffs is governed by the price controls established by the Office, and contained in the *Electricity Distribution Price Determination 2001-2005 – Volume II*. The key provisions of this document define:

- The price control mechanisms applied to determine whether proposed prices are compliant; and
- The constraints on adjusting existing tariffs, called re-balancing constraints.

¹ CitiPower's network tariff schedules are contained in Appendix 1 of this document.

2.1.1 The Price Control

The Office has introduced a tariff based approach to regulate Distribution prices which is essentially a CPI-X price control based on tariff components weighted by their respective component quantities. The price regulation mechanism is based entirely on actual sales and does not rely on forecasts.²

- The maximum $(1+CPI_t)*(1-X_t)*(1+St)$ ratio allowed for 2004 is 1.019905 based on:-
- All Groups Index, Weighted Average of Eight Capital Cities, September quarter 2004 of 142.1
- All Groups Index, Weighted Average of Eight Capital Cities, March quarter 2002 of 138.5
- X_t for 2004 of .01
- St for 2004 of .004108

2.1.2 Re-balancing Constraints

The average price of a network tariff is limited by the re-balancing constraint. This control effectively limits the rate of increase of individual network tariffs. The re-balancing constraint for 2004 for both transmission and distribution tariffs is CPI plus 2%.

Due to large price increases from Vencorp, the re-balancing control on transmission tariffs has been increased to CPI plus 20% for 2004. Despite this increase the constraint prevented CitiPower from pricing transmission charges to fully recover expected transmission costs. Any under collection will be added to the 2005 tariffs through the correction factor.

2.2 Inter Distribution System Charges

The CitiPower distribution network is interconnected with the neighbouring distribution businesses of AGL and United Energy. These interconnected DBs buy and sell distribution capacity across their distribution boundaries. This practice is efficient and avoids the unnecessary duplication of assets.

CitiPower is a net purchaser of inter-DB capacity and under the existing pricing regulations these net costs are recovered through the Network Tariffs.

² For a comprehensive description of the price controls see Office's "Electricity Distribution Price Determination 2001-2005, Volume II – Price Controls"

2.3 Transmission Charges

Costs associated with the use of and the connection to the Transmission system are incurred by CitiPower and incorporated into the transmission component of CitiPower's Network Tariffs to be passed through to distribution customers.

The elements of TUoS are: Prescribed Connection Charges, Excluded Transmission Services, Transmission Use of System charges, Equalisation and Inter Distribution System Charges. For a more detailed description of each of these elements please refer to CitiPower's Distribution Price Review Submission of December 1999.

3 PRICING METHODOLOGY FOR TRANSMISSION CHARGES

The transmission charges which CitiPower pays to VENCORP and SPI PowerNet consist of components that can be separated into demand based and energy based components. These components are then recovered on the basis of “user pays”, to the extent allowed by the binding re-balancing constraint discussed in section 2.1, to derive the TUoS price components of the Network Tariffs.

The following elements of the Transmission Charges were assigned to the Demand component:

- Connection charges;
- Maximum Demand Charges;
- Energy Charges;

The remaining elements listed below were allocated to the Energy Component:

- VENCORP Common Service Charge (\$/MW on all energy transmitted); and
- Equalisation Adjustment (a tax which should be recovered in a manner that gives rise to the least possible distortion of economic signals).

From these components a demand and energy price were calculated.

With reference to experience in the Network Planning group and from the End Use Measurement Survey, it was possible to arrive at a series of load factors and peak contribution factors which were used to allocate transmission costs to the tariff groups in proportion to the demand presented at the time of CitiPower’s system peak and the total energy consumed on the basis of the demand and energy prices derived above.³

3.1 *Transmission Charges*

The transmission charges are then established to collect the required revenue on the basis of available metering data. For example, the price for single rate customers is derived by dividing the revenue required by the forecast energy. In the case of two rate customers, the off peak consumption was assigned the Transmission energy price and the peak consumption was used to recover the balance of the required revenues.

For residential customers it can be assumed that there is a reasonable correlation between energy usage and contribution to the local and system network maximum demand. Network tariffs for larger customers contain a demand related component which in part reflects the cost of transmission system usage.

³ The End Use Measurement Study commenced in 1994 as a Victorian Distribution Industry load monitoring study. Smart meters compile load data on a sample of residential, commercial and industrial customers. The study is ongoing and CitiPower continue to participate.

Energy components of transmission charges such as the Common Service Charge and the Equalisation payment have been allocated to energy components of the transmission tariffs including the off-peak components.

4 NETWORK TARIFFS

As required by the Office's Final Determination, CitiPower has produced a schedule of network, distribution and transmission tariffs. The network tariffs are a simple addition of the distribution and transmission components. These tariff schedules are available in Table 1, Table 2 and Table 3. These tariffs recover the distribution revenue requirement (as set out in the Office's Price Determination) plus the assumed transmission and inter distribution business charges.

CitiPower recognises that tariff design ultimately requires compromises to be reached in relation to a range of conflicting objectives such as:

- economic efficiency;
- energy efficiency;
- pricing stability;
- commercial imperatives; and
- customer equity.

CitiPower has considered, and will continue to consider, in an even-handed way, the trade-offs between these objectives, in formulating new tariffs or modifications to existing ones. These objectives are embodied in CitiPower's network charges, discussed in section 5 below.

4.1 List of approved Network Tariffs

Each of CitiPower's network tariffs are examined in the sections below. All prices and annual bill calculations exclude GST.

4.1.1 Residential Single Rate Tariffs

The Residential Single Rate and Residential Single Rate - Bulk tariffs were originally based on retail Maximum Uniform Tariffs (MUTs) for residential customers. The price for the first 1020 kWh per quarter is slightly lower than for the balance.

This inverse block structure has been introduced to the residential network tariff to maintain a positive rudimentary demand management signal to larger than average consumers. The average consumption of CitiPower single rate residential customers is approximately 4300 kWh per annum which is close to the 1020 kWh per quarter (4080 kWh per annum) used in the MUTs GD and GR. For simplicity the same 1020 kWh transition point has been retained.

On the basis that residential customers, in aggregate, present a different demand profile to commercial and industrial customers these tariffs are restricted to residential customers. Figures 1 and 2 below illustrate diagrammatically the impact the significant reductions have had in delivering price reductions to small residential customers, and the block structure in providing a price incentive to large residential customers.

Figure 1: Residential Single Rate – Average Annual Bill by Customer Segment

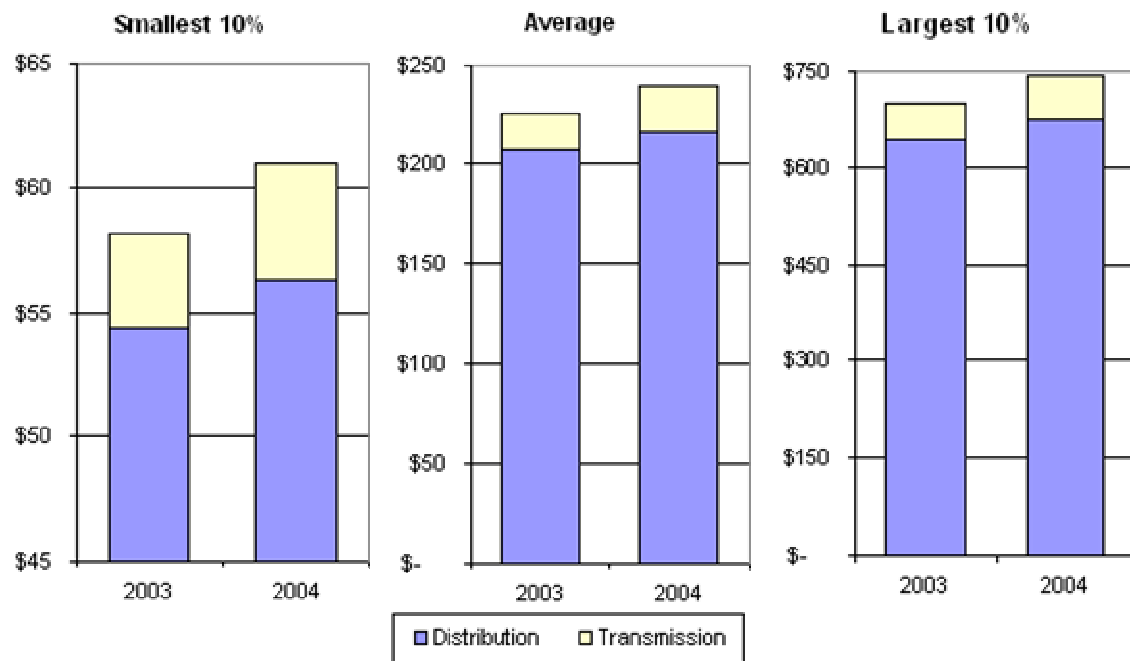
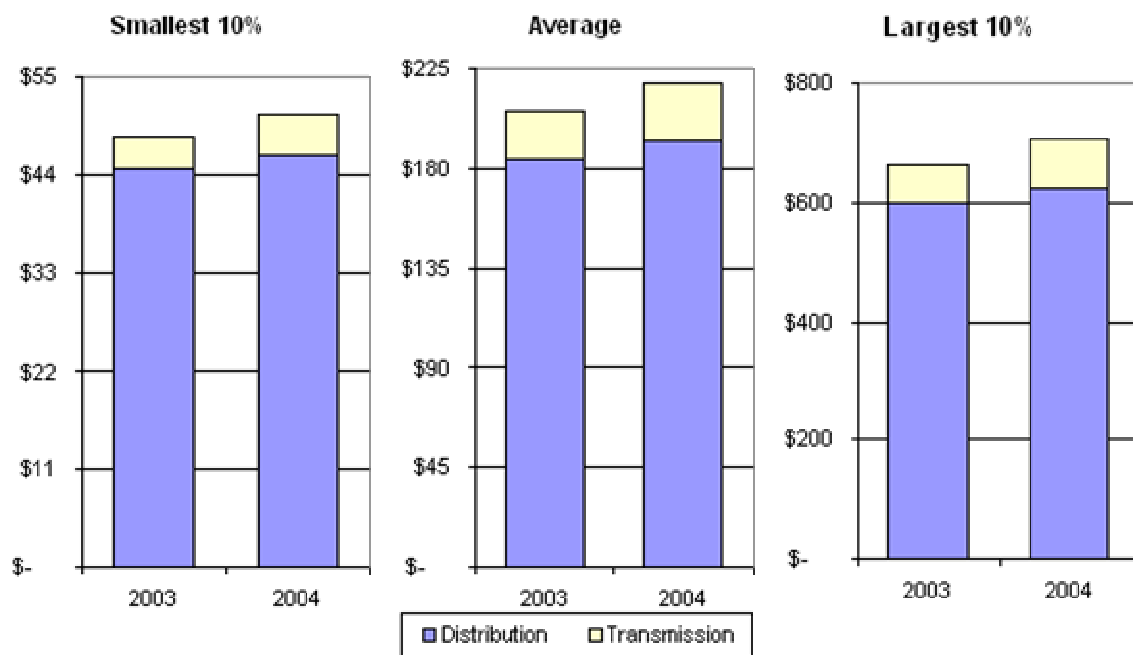


Figure 2: Residential Single Rate Bulk – Average Annual Bill by Customer Segment



4.1.2 Residential Two Rate Tariffs

The Residential Two Rate and Residential Two Rate - Bulk tariffs were originally also based on retail Maximum Uniform Tariffs (MUTs) for residential customers and include a rudimentary demand signal through the differential pricing for time of use.

On the basis that residential customers, in aggregate, present a different demand profile to commercial and industrial customers these tariffs are restricted to residential customers.

The impact of these tariffs is shown diagrammatically in Figures 3 and 4 below.

Figure 3: Residential Two Rate – Average Annual Bill by Customer Segment

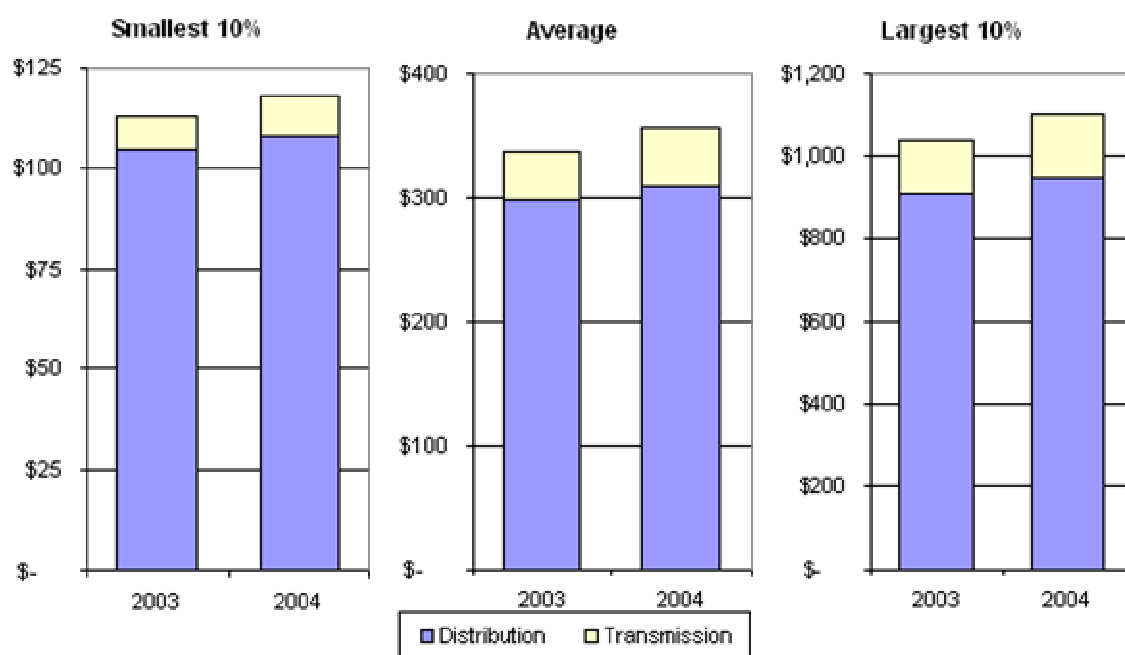
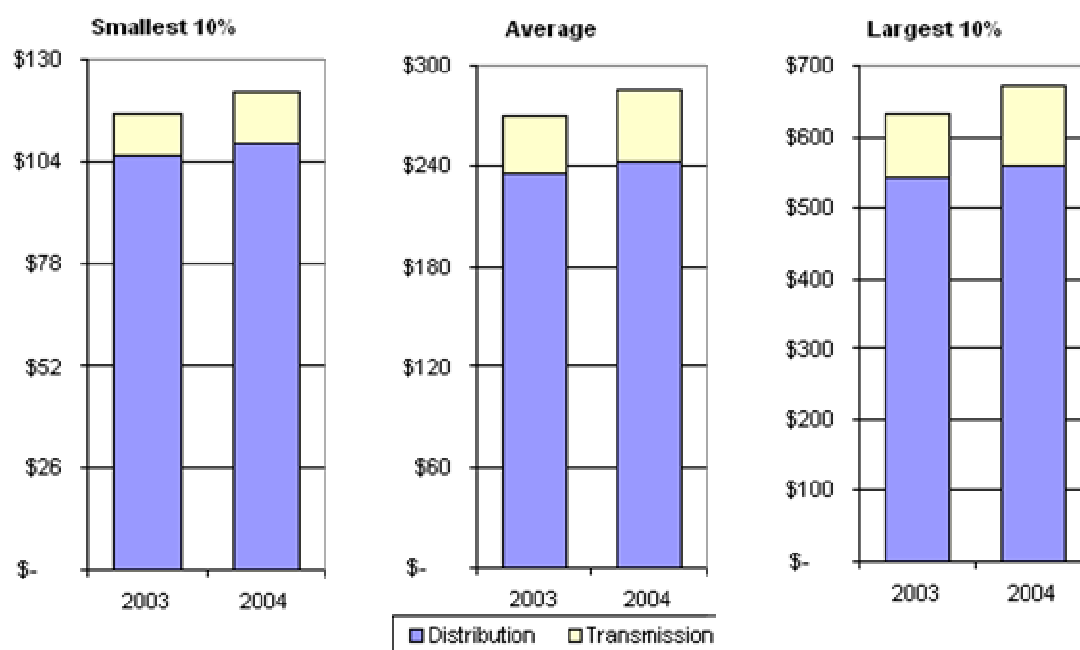


Figure 4: Residential Two Rate Bulk– Average Annual Bill by Customer Segment



4.1.3 Dedicated Supplies (Storage Water & Space Heating)

These tariffs are available for controlled loads such as storage water heating and space heating for residential purposes only. Being controlled loads, they make no contribution to peak demands on either the transmission or distribution system.

These tariffs currently make a substantial contribution to the total revenue without driving capacity increase while at the same time facing intense competition from alternative fuel sources such as natural gas. Therefore the large reductions in the 2001 rates to make these tariffs more affordable have been maintained. This should help discourage these sales being unnecessarily eroded to the detriment of system utilisation and therefore overall economic efficiency of the distribution network.

Figure 5: Dedicated Supplies– Average Annual Bill by Customer Segment

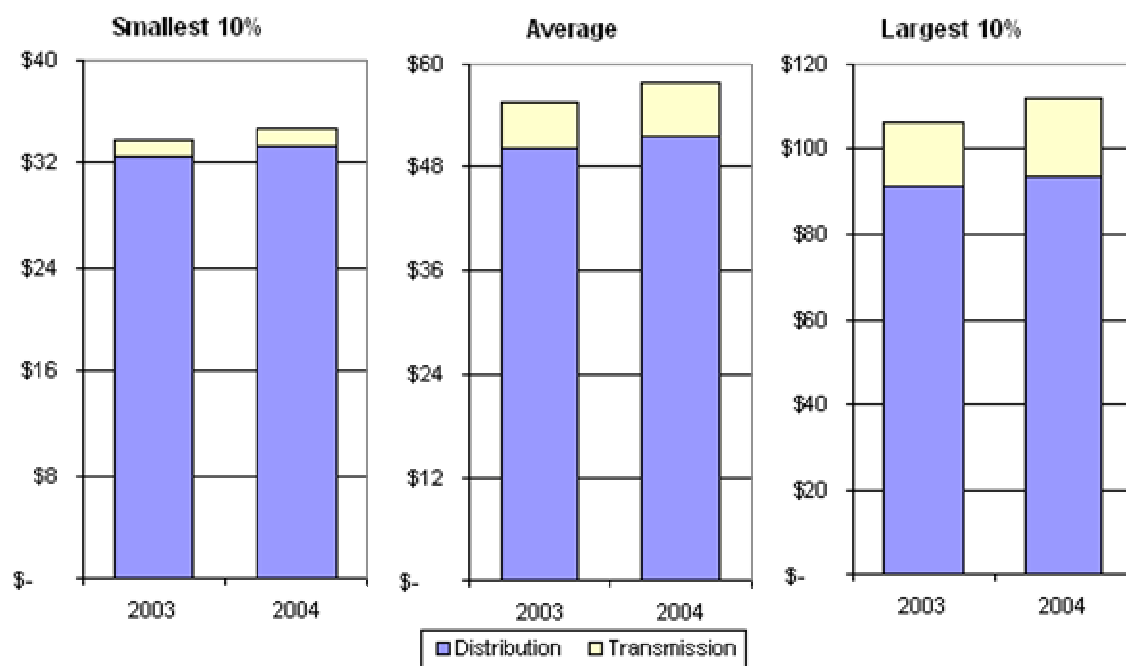
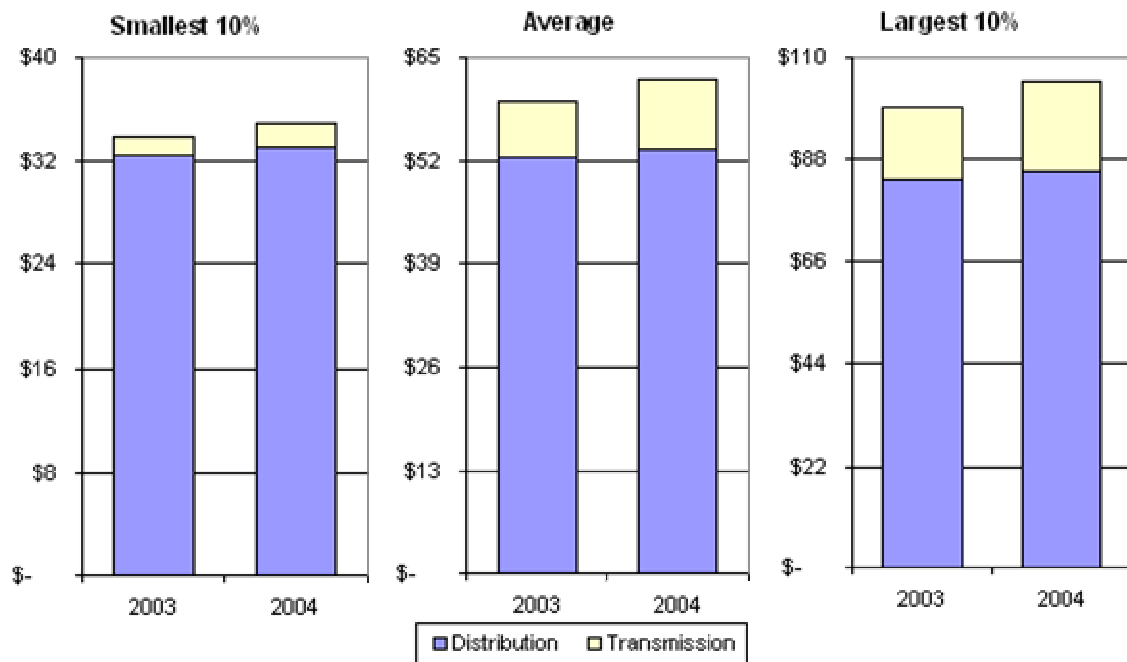


Figure 6: Dedicated Supplies Bulk– Average Annual Bill by Customer Segment



4.1.4 General Purpose Single Rate .

These tariffs are available to small and medium business and industrial customers.

No major changes have been made to the structure of these tariffs and new customers are encouraged to consider the two rate or maximum demand tariffs as an alternative, as they promote more efficient utilisation of the network by providing a time of use signal.

Figure 7: General Purpose Single Rate – Average Annual Bill by Customer Segment

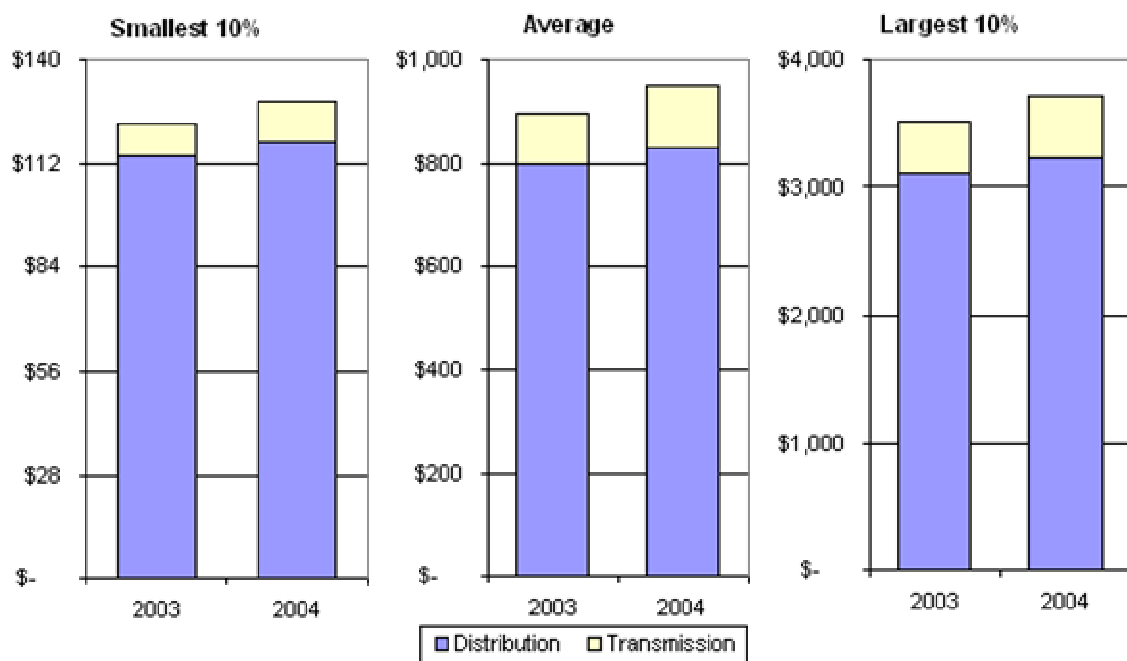
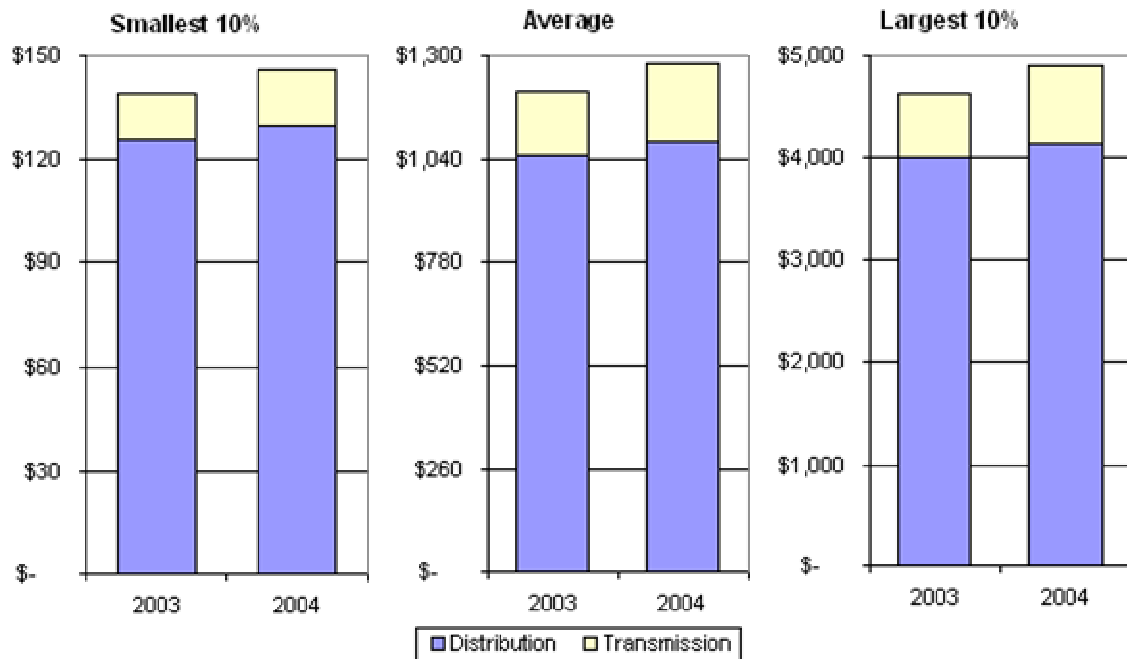


Figure 8: General Purpose Single Rate Bulk – Average Annual Bill by Customer Segment



4.1.5 5.1.5 General Purpose Two Rate (5 day peak) .

These tariffs were based on the two rate General Purpose MUT used by small and medium business and industrial customers and include a rudimentary demand signal through the differential pricing for time of use.

Figure 9: General Purpose Two Rate (5 day peak) – Average Annual Bill by Customer Segment

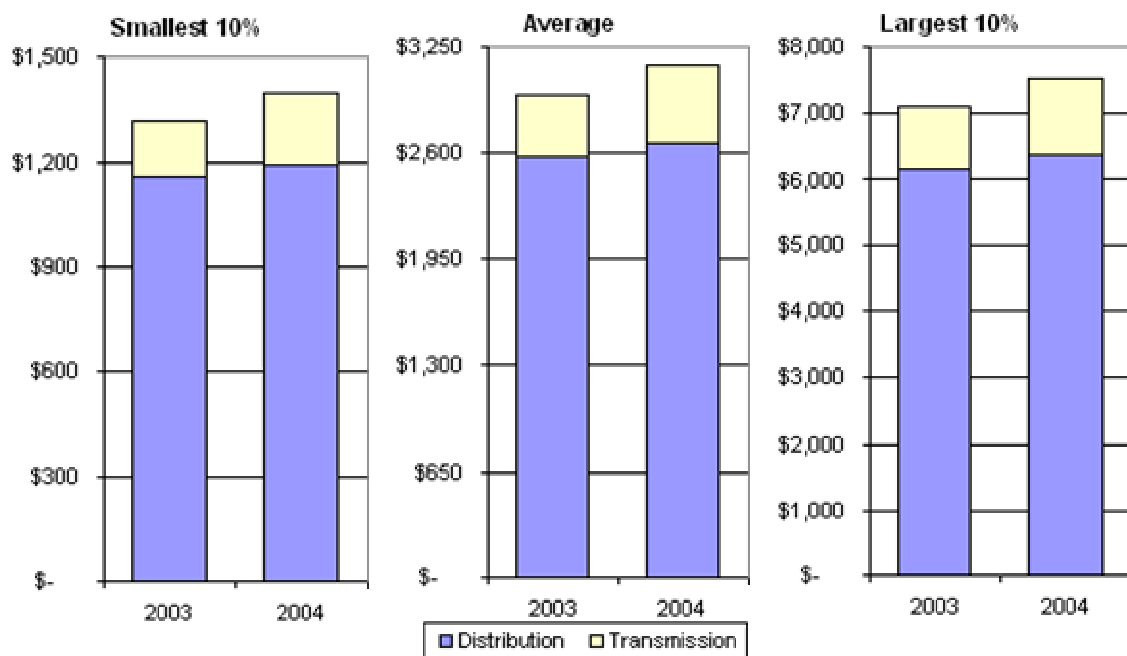
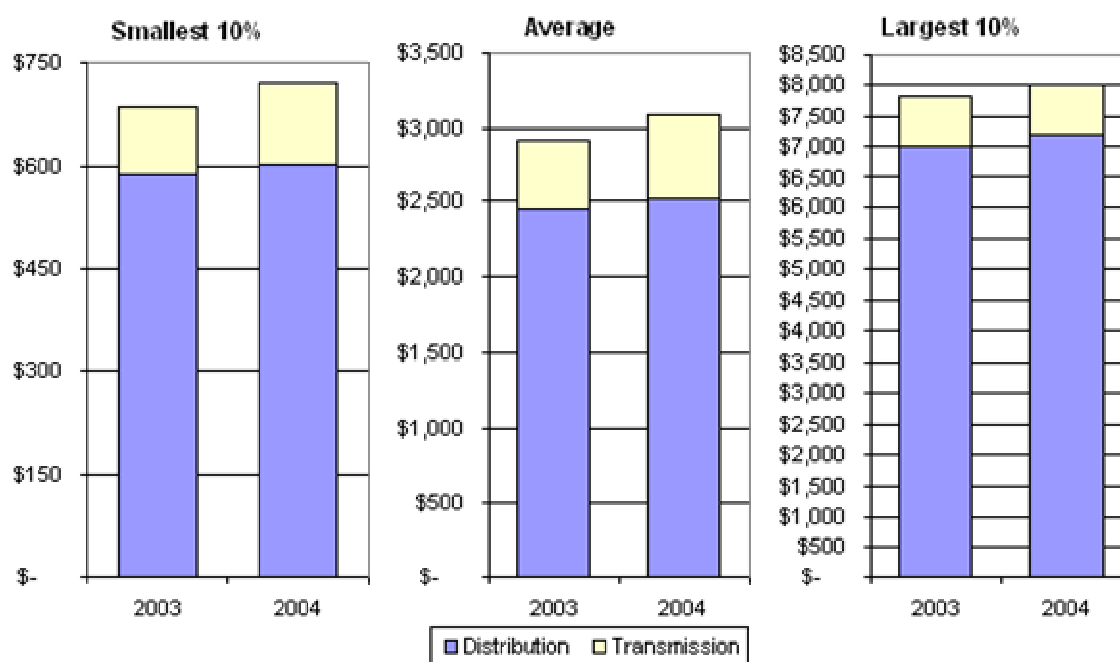


Figure 10: General Purpose Two Rate (5 day peak) Bulk – Average Annual Bill by Customer Segment



4.1.6 General Purpose Two Rate (7 day peak)

These tariffs were based on the obsolete MUT E1, and are available only to existing installations supplied under this tariff. Existing customers are encouraged to consider the two rate, 5 day peak (weekends off-peak) or maximum demand tariffs as an alternative because they promote more efficient utilisation of the network by providing an appropriate time of use signal.

Figure 11: General Purpose Two Rate (7 day peak) – Average Annual Bill by Customer Segment

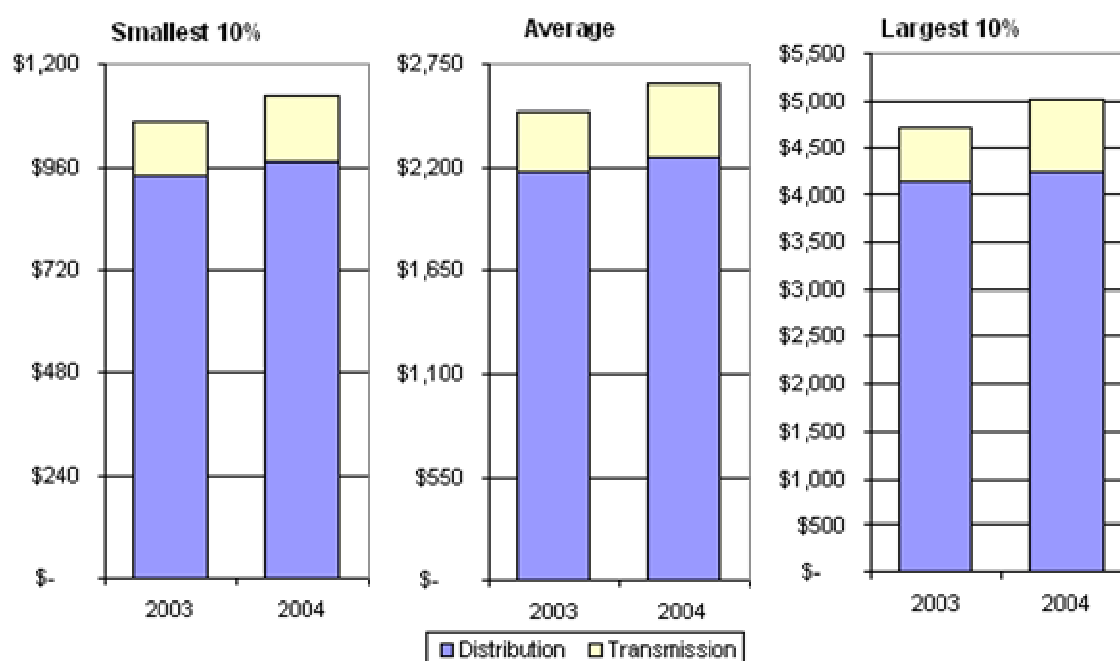
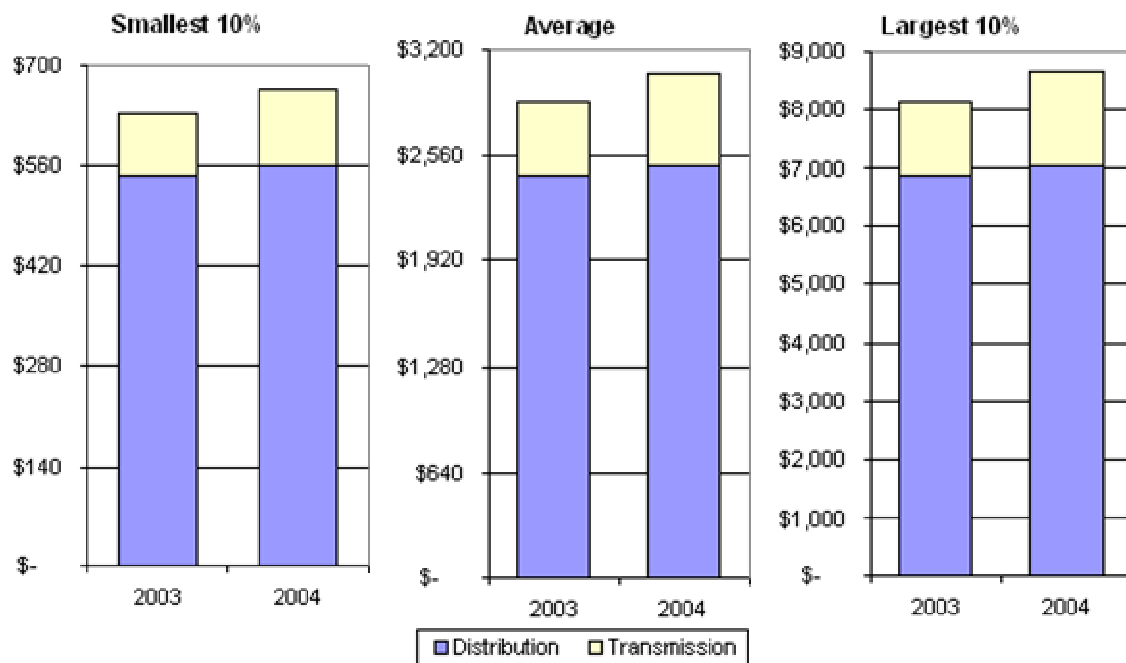


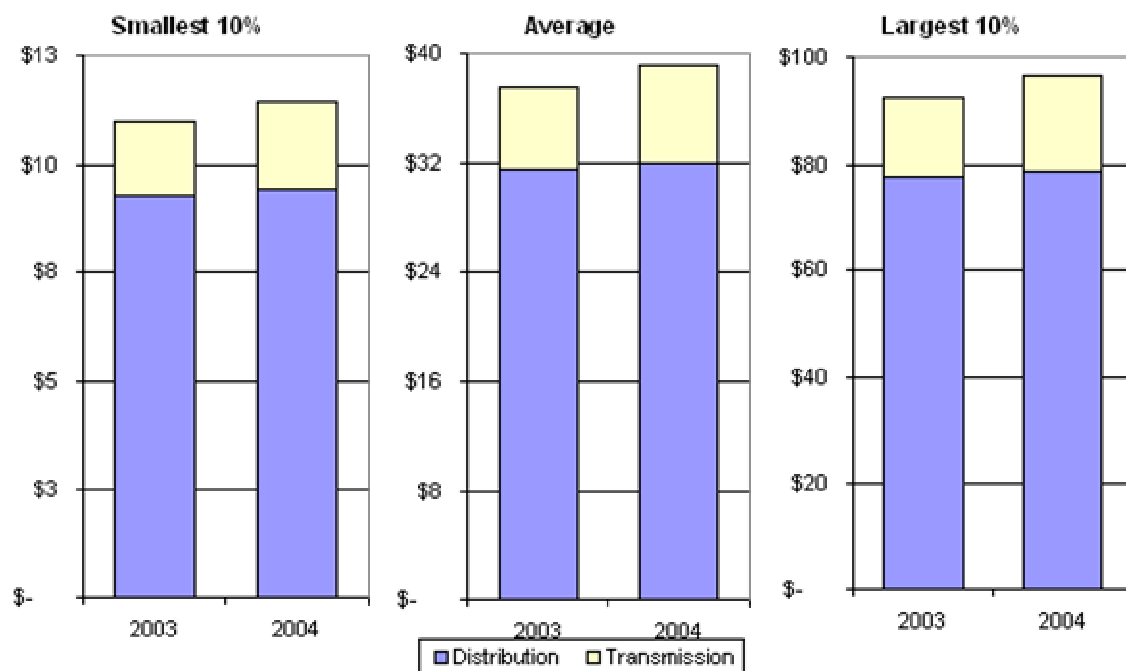
Figure 12: General Purpose Two Rate (7 day peak) Bulk – Average Annual Bill by Customer Segment



4.1.7 Large Two Rate (7 day peak)

This tariff is based on the obsolete MUT E1 and is available only to existing installations supplied under this tariff. There are relatively few customers supplied under this tariff and they are encouraged to consider the Large LV Demand metered tariff as an alternative because it promotes more efficient utilisation of the network by providing an appropriate time of use signal.

Figure 13: Large Two Rate (7 day peak) – Average Annual Bill (thousands of dollars) by Customer Segment

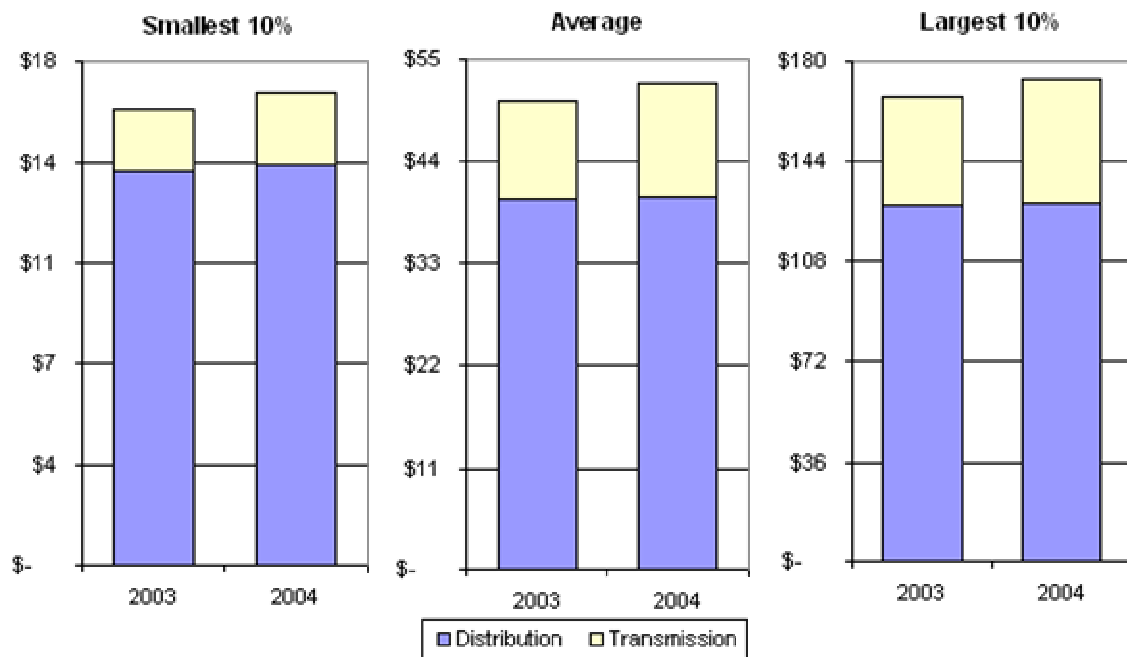


4.1.8 Large LV Demand Metered

This tariff is designed for larger commercial and industrial customers and provides both a demand signal and a time of use signal to promote efficient network use.

The calculation of the demand component is based on a “contract demand” which sets minimum billed demand for at least one billing year.

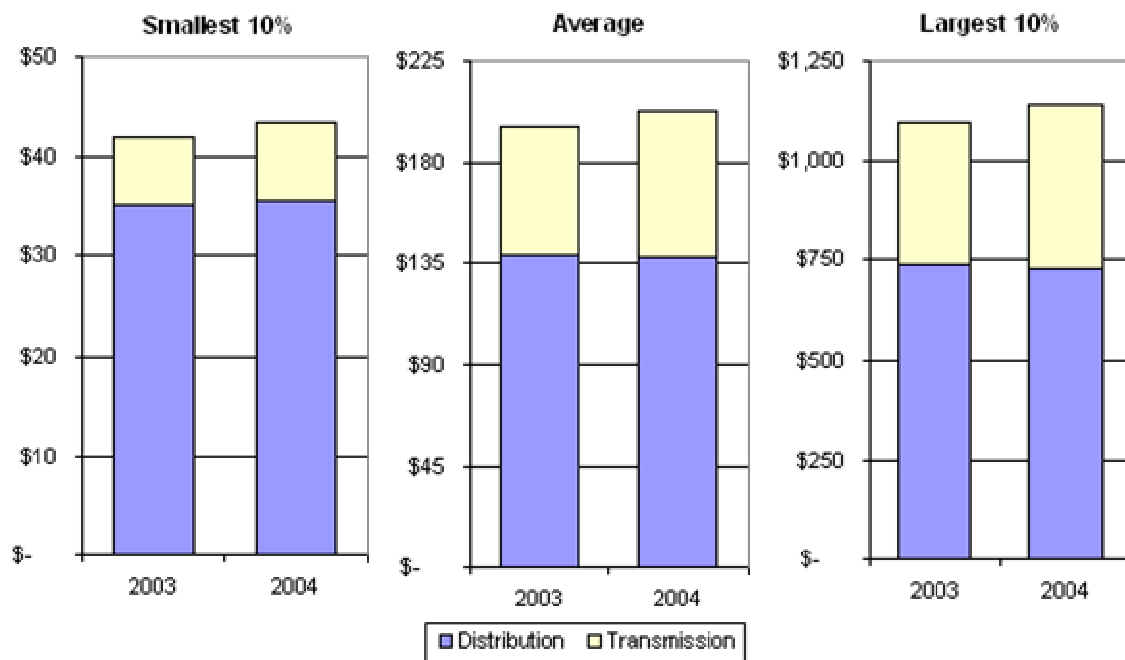
Figure 14: Large LV Demand Metered (Bulk) – Average Annual Bill (thousands of dollars) by Customer Segment



4.1.9 HV Demand Metered

This tariff is designed for larger commercial and industrial customers who require high voltage supply. It provides both a demand signal and a time of use signal to promote efficient network use. As with the Large LV Demand Metered Tariff, the calculation of the demand component is based on a “contract demand”.

Figure 15: HV Demand Metered – Average Annual Bill (thousands of dollars) by Customer Segment



4.1.10 HV Demand Metered (D1)

This tariff is designed for larger commercial and industrial customers who require high voltage supply (with a total maximum demand greater than 40,000kW) who are connected at 11/6.6kV via 2 or more dedicated closed ring underground feeders without directional powerflow control and without auto/transfer capability. It provides both a demand signal and a time of use signal to promote efficient network use.

4.1.11 Subtransmission

This tariff applies only to a very limited number of large connections operating at 22,000 volts. It is heavily dominated by the transmission component

4.1.12 Unmetered Supplies

This tariff applies only to those situations where it is not economic to install a meter such as public lighting, telephone cabinets etc.



APPENDIX 1 : DISTRIBUTION TARIFF SCHEDULE (GST EXCLUSIVE)

1 JANUARY 2004 – 31 DECEMBER 2004

Table 1 – Distribution tariffs effective 1 January 2004

DUoS Tariff	Code	Standing charges \$/cust/pa	Demand charges \$/kW/pa	Minimum Demand kW	Peak charges c/kWh		Peak charges c/kWh	Off peak charges c/kWh
					First 1020kWh/qtr	Balance		
Residential Single Rate	C1R	21.790			4.136	5.427		
Residential Single Rate - Bulk	C1RB	18.790			3.432	4.359		
Residential Two Rate	C2R	48.650					5.438	1.247
Residential Two Rate - Bulk	C2RB	47.960					4.491	1.027
Dedicated Supplies	CDS	28.100						0.773
Dedicated Supplies - Bulk	CDSB	27.760						0.654
General Purpose Single Rate	C1G	49.080					5.073	
General Purpose Single Rate - Bulk	C1GB	41.860					4.186	
General Purpose Two Rate (5 day peak)	C2G5	116.560					5.965	1.375
General Purpose Two Rate (5 day peak) - Bulk	C2G5B	108.670					4.654	1.057
General Purpose Two Rate (7 day peak)	C2G7	105.650					5.010	1.368
General Purpose Two Rate (7 day peak) - Bulk	C2G7B	97.270					3.903	1.047
Unmetered Supplies	C2U						5.840	1.428
Large Two Rate (7 day peak) - Bulk	C2L7	102.240					4.325	1.284
Large LV Demand Metered	C2DL	3,847.640	62.850	120 kW			0.802	0.820
Large LV Demand Metered - Bulk	C2DL-B	3,847.640	57.500	120 kW			0.498	0.806
HV Demand Metered	C2DH	2,510.060	43.670	1,000 kW			0.300	0.450
HV Demand Metered (D1)	C2DHD1	2,462.160	24.930	40,000 kW			0.069	0.064
Subtransmission	C2DT	2,530.060	9.660	10,000 kW			0.086	0.130

Some Tariffs apply to specific customer groups in specific areas within the CitiPower distribution network only. CitiPower will provide specific information on request.



APPENDIX 1 : TRANSMISSION TARIFF SCHEDULE (GST EXCLUSIVE)

1 JANUARY 2004 – 31 DECEMBER 2004

Table 2 – Transmission tariffs effective 1 January 2004

TUoS Tariff	Code	Standing charges \$/cust/pa	Demand charges \$/kW/pa	Minimum Demand kW	Peak charges c/kWh		Peak charges c/kWh	Off peak charges c/kWh
					First 1020kWh/qtr	Balance		
Residential Single Rate	C1R				0.554	0.554		
Residential Single Rate - Bulk	C1RB				0.554	0.554		
Residential Two Rate	C2R						0.972	0.222
Residential Two Rate - Bulk	C2RB						0.972	0.222
Dedicated Supplies	CDS							0.221
Dedicated Supplies - Bulk	CDSB							0.222
General Purpose Single Rate	C1G						0.773	
General Purpose Single Rate - Bulk	C1GB						0.773	
General Purpose Two Rate (5 day peak)	C2G5						1.138	0.222
General Purpose Two Rate (5 day peak) - Bulk	C2G5B						1.139	0.221
General Purpose Two Rate (7 day peak)	C2G7						0.908	0.222
General Purpose Two Rate (7 day peak) - Bulk	C2G7B						0.908	0.222
Unmetered Supplies	C2U						0.224	0.221
Large Two Rate (7 day peak) - Bulk	C2L7						0.998	0.214
Large LV Demand Metered	C2DL		5.960	120 kW			0.934	0.214
Large LV Demand Metered - Bulk	C2DL-B		5.860	120 kW			0.919	0.210
HV Demand Metered	C2DH		4.970	1,000 kW			1.015	0.207
HV Demand Metered (D1)	C2DHD1		6.920	40,000 kW			0.973	0.316
Subtransmission	C2DT		2.000	10,000 kW			1.133	0.192

Some Tariffs apply to specific customer groups in specific areas within the CitiPower distribution network only. CitiPower will provide specific information on request.



APPENDIX 1 : NETWORK TARIFF SCHEDULE (GST EXCLUSIVE)

1 JANUARY 2004 – 31 DECEMBER 2004

Table 3 – Network tariffs effective 1 January 2004

NUoS Tariff	Code	Standing charges \$/cust/pa	Demand charges \$/kW/pa	Minimum Demand kW	Peak charges c/kWh		Peak charges c/kWh	Off peak charges c/kWh
					First 1020kWh/qtr	Balance		
Residential Single Rate	C1R	21.790			4.690	5.981		
Residential Single Rate - Bulk	C1RB	18.790			3.986	4.913		
Residential Two Rate	C2R	48.650					6.410	1.469
Residential Two Rate - Bulk	C2RB	47.960					5.463	1.249
Dedicated Supplies	CDS	28.100						0.994
Dedicated Supplies - Bulk	CDSB	27.760						0.876
General Purpose Single Rate	C1G	49.080					5.846	
General Purpose Single Rate - Bulk	C1GB	41.860					4.959	
General Purpose Two Rate (5 day peak)	C2G5	116.560					7.103	1.597
General Purpose Two Rate (5 day peak) - Bulk	C2G5B	108.670					5.793	1.278
General Purpose Two Rate (7 day peak)	C2G7	105.650					5.918	1.590
General Purpose Two Rate (7 day peak) - Bulk	C2G7B	97.270					4.811	1.269
Unmetered Supplies	C2U						6.064	1.649
Large Two Rate (7 day peak) - Bulk	C2L7	102.240					5.323	1.498
Large LV Demand Metered	C2DL	3,847.640	68.810	120 kW			1.736	1.034
Large LV Demand Metered - Bulk	C2DL-B	3,847.640	63.360	120 kW			1.417	1.016
HV Demand Metered	C2DH	2,510.060	48.640	1,000 kW			1.315	0.657
HV Demand Metered (D1)	C2DHD1	2,462.160	31.850	40,000 kW			1.042	0.380
Subtransmission	C2DT	2,530.060	11.660	10,000 kW			1.219	0.322

Some Tariffs apply to specific customer groups in specific areas within the CitiPower distribution network only. CitiPower will provide specific information on request.

APPENDIX 1 : ADDITIONAL TERMS

The following terms apply:

- (a) Large Low Voltage Tariffs have a nominal voltage of less than 1000 volts.
- (b) High Voltage Tariffs have a nominal voltage of greater than and equal to 1000 volts and less than and equal to 22,000 volts.
- (c) Sub-Transmission Tariffs have a nominal voltage equal to 22,000 volts.
- (d) Tariff HV Demand Metered (D1) applies to customers (with a maximum demand greater than 40,000kW) who are connected at 11/6.6kV via 2 or more dedicated closed ring underground feeders without directional powerflow control and without auto/transfer capability.
- (e) Tariffs designated "Bulk" apply to customers supplied directly from on site substation terminals where there are no Citipower low voltage distribution assets beyond the substation.